

JORDI BUILS

P.h.D in Computational Chemistry

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EXPERIENCE

Postodctoral Researcher

Prof. Carles Bo Jané - Institute of Chemical Research of Catalonia (ICIQ)

📅 December 2024 - Currently

📍 Tarragona, Spain

Substitute Professor

Universitat Autònoma de Barcelona (UAB)

📅 September 2025 - Currently

📍 Cerdanyola del Vallès, Spain

EDUCATION

P.h.D Degree in Chemical science and Technology (Cum Laude and International Mention)

Prof. Carles Bo Jané and Dr. Mireia Segado Centellas at Institute of Chemical Research of Catalonia (ICIQ)

📅 November 2021 - December 2024

📍 Tarragona, Spain

Master Degree in Chemical Science and Technology

Prof. Àngel Terrón Homar - University of the Balearic Islands

📅 September 2020 - September 2021

📍 Mallorca, Spain

Bachelor Degree in Chemistry

Prof. Àngel Terrón Homar - University of the Balearic Islands

📅 September 2015 - February 2020

📍 Mallorca, Spain

Skills

- **Density Functional Theory code:** ADF (Amsterdam Density Functional, AMS (Amsterdam Modeling Suite) Gaussian
- Programming Languages: Python, Bash
- Libraries: Scipy, Numpy, Pandas, Scikit-Learn, PyTorch
- Open data: Git (GitLab and GitHub), ioChem-BD
- Languages: Catalan, Spanish, and English

Open Sourced projects

POMSimulator 1.0.0 : Chemistry toolkit for simulating the aqueous speciation of metal-oxo clusters

🔗 <https://github.com/petrusen/pomsimulator>

POMSimulator 2.0.0: Chemistry toolkit for simulating the aqueous speciation of metal-oxo clusters

🔗 <https://github.com/petrusen/pomsimulator>

POMs-GibbsNet : A DNN model for the prediction of thermodynamic properties of metal-oxo clusters

♥ <https://gitlab.com/jbuils/poms-gibbsnet>

POMs DataBase: A collection of metal-oxo clusters for the training of GibbsNet

🔗 [ioChem-bd](#)

Publications

- L. Junkers, D. Garay-Ruiz, **Jordi Buils**, R. Silberg, G. B. Strapasson, K. Jensen, C. Bo. "Uncovering polyoxometalate speciation in hydrothermal systems by combining computational simulation with X-ray total scattering." *Journal of the American Chemical Society*, (2025)
- **J. Buils**, A. Terron, M. Barceló-Oliver, J.J. Fiol, A. García-Raso, R.M. Gomila, A. Frontera. "Synthesis, X-ray characterization, and DFT calculations of gold-nucleobase complexes: On the importance of regium bond and anion- π interactions." *CystEngComm*, (2025)
- F. Ibrahima Bamba, C. Falaise, N. Leclerc, M. Haouas, G. Gbassi, P. Atheba, M. Fregnaux, **J. Buils**, M. Segado-Centellas, C. Bo, E. Cadot. "Capping the Electronic Lone Pair of the As(III) Central Atom in the Keggin-type Anion: From Experimental-Theoretical Interplay to Evidence." *Inorganic Chemistry* (2025)
- J. Bustos, M. Shohel, A. G. Buzanich, L. Zakharov, **J. Buils**, M. Segado-Centellas, C. Bo, M. Nyman. "Technetium and Rhenium Auto-reduction, Polymerization and Lability towards Group VII Polyoxometalate Chemistry." *Chemistry-A European Journal* 31, 21 (2025)
- **J. Buils**, D. Garay-Ruiz, E. Petrus, M. Segado-Centellas, C. Bo. "Towards a universal scaling method for predicting equilibrium constants of polyoxometalates." *Digital Discovery*, 4, 4, 970-978 (2025)
- D. Garay-Ruiz, **J. Buils**, N. A. G. Bandeira, S. Floquet, E. Cadot, C. Bo. "Thinking about the Box: Exploring the Electronic Structure of Cuboidal-Shaped Mo_{84} Anionic Nanocapsule." *Inorganic Chemistry* (2025)
- **J. Buils**, D. Garay-Ruiz, E. Petrus, M. Segado-Centellas, C. Bo. "Computational insights into aqueous speciation of metal-oxide nanoclusters: an in-depth study of the Keggin phosphomolybdate." *Chemical Science*, 15, 35, 14218-14227 (2024)
- E. Petrus, **J. Buils**, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. "POMSimulator: An open-source tool for predicting the aqueous speciation and self-assembly mechanisms of polyoxometalates." *Journal of Computational Chemistry*, 45, 26, 2242-2250 (2024)

- D. Cebotari, **J. Buils**, O. Garbuz, G. Balan, J. Marrot, V. Guérineau, D. Touboul, M. Haouas, M. Segado-Centellas, C. Bo, A. Gulea, S. Floquet. "A new series of Mo(V)₂O₂S₂-based thiosemicarbazone complexes: Solution and DFT studies, and antifungal and antioxidant activities." *Journal of Inorganic Biochemistry*, 245 (2023)
- M. De las Nieves-Piña, S. Burguera, **J. Buils**, M. A. Crespi, J. E. Morales, J. Pons, A. Bauzá, A. Frontera. "Substituent Effects in π -Hole Regium Bonding Interactions Between Au(p-X-Py)₂ Complexes and Lewis Bases: An ab initio Study." *ChemPhysChem*, 23, 8 (2022)
- A. Terrón, **J. Buils**, T. J. Mooibroek, M. Barceló-Oliver, A. García-Raso, J.J. Fiol, A. Frontera "Synthesis, X-ray characterization and regium bonding interactions of a trichlorido (1-hexylcytosine) gold(III) complex." *Chemical Communications*, 56, 24, 3524-3527 (2020)

Conferences

- ✿ **J. Buils**, D. Garay-Ruiz, E. Petrus, M. Segado-Centellas, D. Balcells, C. Bo. (2025) "Data-driven strategies for the study of the aqueous speciation of polyoxometalates using POMSimulator", Poster communication at FMOCS25, Mainz, Germany
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, D. Balcells, C. Bo. (2025) "Data-Driven Methods for Solving Multi-Species Multi-Equilibria Self-Assembly of Metal-Oxide Nanoclusters" Oral communication at WATOC25, Oslo, Norway
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2025) "Predicció de l'especiació aquosa de heteropolioxometal·lats: el fosfomolibdat i l'arsenomolibdat." Oral communication (flash) at "2^a Reunió de Química Inorgànica i Organometàl·lica de la SCQ", Tarragona, Spain
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2024) "DFT Simulations of Complex Reaction Networks to solve the aqueous speciation of polyoxometalates" Oral communication (flash) + poster at "EuChemS 2024", Dublin, Ireland
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2024) "DFT Simulations of Complex Reaction Networks to solve the aqueous speciation of polyoxometalates" Oral communication (flash) at "ESPA 2024", Tarragona, Spain
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2024) "Simulacions de Xarxes de Reacció Complexes d'Òxids de Molibdè." Oral communication (flash) at "13^a Trobada de Joves Investigadors dels Països Catalans", Tarragona, Spain
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2023) "DFT Simulations of Complex Reaction Networks: Aqueous speciation of Molybdenum Oxides and Formation of the Keggin anion." Poster communication at "EuChemS CompChem 2023", Thessaloniki, Greece
- ✿ **J. Buils**, E. Petrus, D. Garay-Ruiz, M. Segado-Centellas, C. Bo. (2023) "DFT Simulations of Complex Reaction Networks: Aqueous speciation of Molybdenum Oxides and Formation of the Keggin anion." Oral communication at "XXXIX Reunión Bienal de la Sociedad Española de Química", Zaragoza, Spain
- ✿ **J. Buils**, E. Petrus, M. Segado-Centellas, C. Bo. (2023) "Understanding the aqueous speciation of heteropolioxometalates with POMSimulator: the Keggin anion." Oral communication (flash) at FMOCS23, Tarragona, Spain
- ✿ **J. Buils**, E. Petrus, M. Segado-Centellas, C. Bo. (2023) "POMSimulator: Estudiant el mecanisme de formació de l'anió Keggin en solució aquosa" Poster communication at "1^a Reunió de Química Teòrica i Computacional de la SCQ", Barcelona, Spain
- ✿ **J. Buils**, E. Petrus, M. Segado-Centellas, C. Bo. (2022) "On the use of the POMSimulator to understand the formation mechanism of metal oxide nanoclusters: the Keggin anion" Poster communication at "VI PhD Day", Tarragona, Spain
- ✿ **J. Buils**, E. Petrus, M. Segado-Centellas, C. Bo. (2022) "On the use of the POMSimulator to understand the formation mechanism of metal oxide nanoclusters: the Keggin anion" Poster communication at "VI PhD Day", Rimini, Italy
- ✿ **J. Buils**, E. Petrus, M. Segado-Centellas, C. Bo. (2022) "On the use of the POMSimulator to understand the formation mechanism of metal oxide nanoclusters: the Keggin anion" Poster communication at "Summer School in Polyoxometalates Chemistry for Fundamentals and Applications", La Rochelle, France

Workshops and courses

- ✿ Using the Amsterdam Modeling Suite in HPC Systems. Organized by SCM AMS **(2022)**
Online 16h
- ✿ Skibotn hackathon at Trömsö **(2024)** presential 4 days

Supervision and mentoring

- ✿ Mentoring of Dr. L. Susanne Junkers. University of Copenhagen
During three months of her P.h.D secondment, supervised her use of POMSimulator to adapt it to thermosynthesis conditions.
(Jan 2024 - Apr 2024)
- ✿ Supervision of Ms Farzaneh Hosseini. Institute of Chemical Research of Catalonia
Since start of her P.h.D, supervised her adaptation to the group working line.
Application of POMSimulator to the speciation of polyoxometalates in proteins and DFT studies of rare metal polyoxometalates (Tc, Re, Ln, Pm, Lu ...)
(2025-currently)
- ✿ Supervision of Mr Stefan Jordi Giurgiu. Institute of Chemical Research of Catalonia
Programa Joves i Ciència from Fundació La Pedrera, Estades de Grau
Study of Uranyl speciation in carbonate solution conditions using POMSimulator
(Jul 2025 - Sep 2025)